

Assignment 12

2303A510G3

Batch-03

CODE:

```
import java.util.*;
```

```
public class Main {
```

```
    public static int maxDifficulty(int[] marks, int[] diff, int limit, int[][] bonusPairs, int[] bonusVal)
    {
        int n = marks.length;
        int maxMask = 1 << n;
        int result = 0;
```

```
        for (int mask = 0; mask < maxMask; mask++) {
            int totalMarks = 0;
            int totalDiff = 0;
            for (int i = 0; i < n; i++) {
                if ((mask & (1 << i)) != 0) {
                    totalMarks += marks[i];
                    totalDiff += diff[i];
                }
            }
            if (totalMarks <= limit) {
                for (int i = 0; i < bonusPairs.length; i++) {
                    int a = bonusPairs[i][0];
                    int b = bonusPairs[i][1];
                    if ((mask & (1 << a)) != 0 && (mask & (1 << b)) != 0) {
                        totalDiff += bonusVal[i];
                    }
                }
                result = Math.max(result, totalDiff);
            }
        }
        return result;
    }
```

```
    public static int maxAccuracy(int[] acc, int[] cost, int limit) {
        int n = acc.length;
        int maxMask = 1 << n;
        int result = 0;
```

```
        for (int mask = 0; mask < maxMask; mask++) {
            int totalCost = 0;
            int totalAcc = 0;
            for (int i = 0; i < n; i++) {
                if ((mask & (1 << i)) != 0) {
                    totalCost += cost[i];
                    totalAcc += acc[i];
                }
            }
        }
    }
```

```

        }
    }
    if (totalCost <= limit) {
        result
    = Math.max(result, totalAcc);
    }
    }
    return result;
}

    public static void main(String[] args) {
int[] marks = {5,4,6,3};    int[] diff = {8,6,7,5};
int limit1 = 10;    int[][] bonusPairs = {{0,2}};
int[]
bonusVal = {3};
    System.out.println(maxDifficulty(marks, diff, limit1, bonusPairs, bonusVal));
    int[] acc =
{4,5,3,6};
int[] cost = {3,4,2,5};    int limit2
= 8;
    System.out.println(maxAccuracy(acc, cost, limit2));
    }
}

```

OUTPUT:

14
10